

POKHARA UNIVERSITY

Level: Bachelor
Semester: Fall
Programme: BE
Course: Data Structure and Algorithm

Year : 2020
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

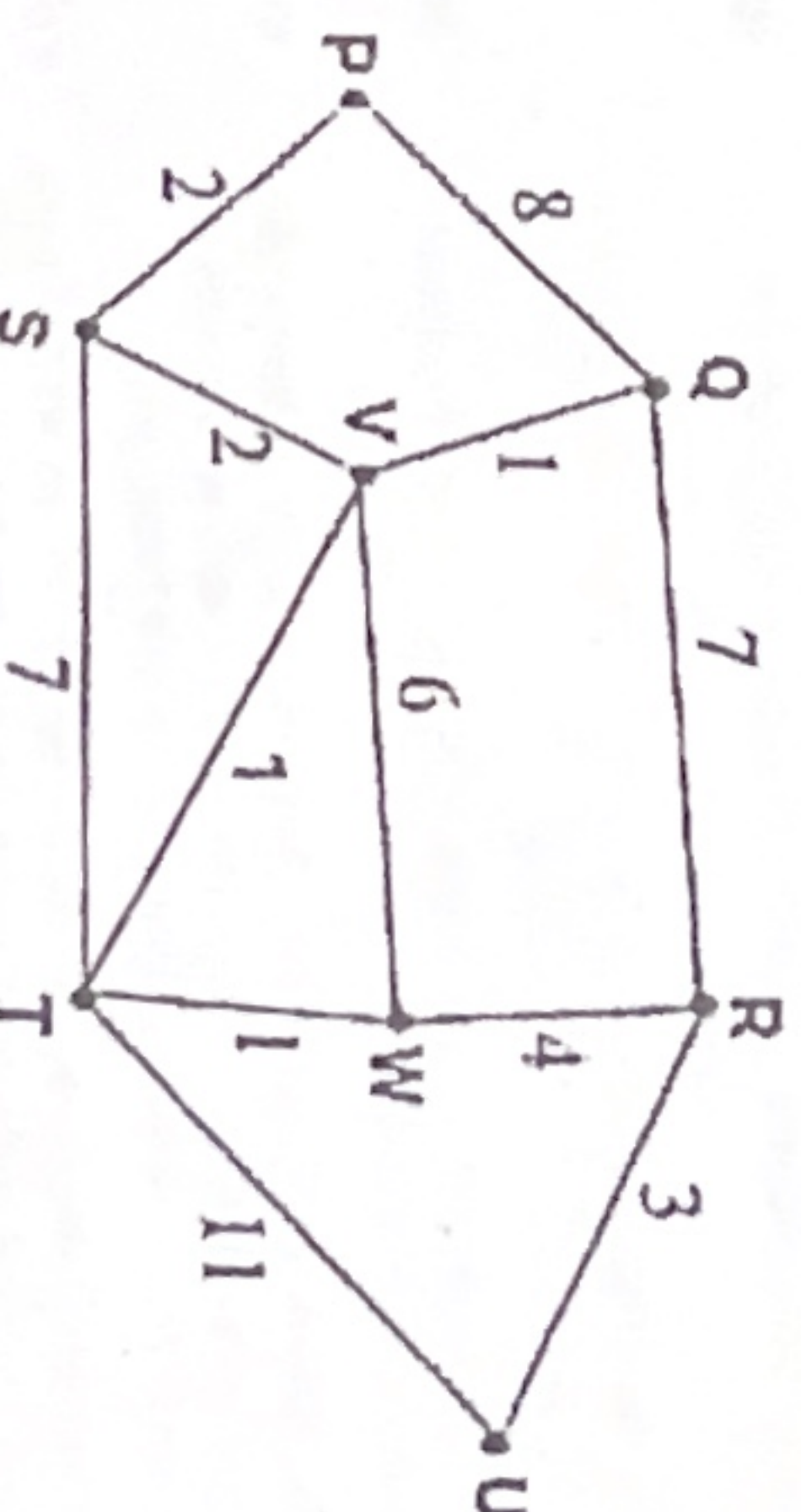
The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Why do you need data structure? Data Structure is an abstract data type. Do you agree with this Statement? If so, explain with example. 7
- b) Why do you convert infix expression to postfix expression? Explain how do you use stack to convert the following infix expression into postfix: $A+(B*(C-(D/E^{\wedge}F)*G)*H$ Where $\wedge$ is power operator. 8
2. a) What are the shortcomings of linear queue? How are they solved in circular queue? Write algorithms for enqueue and dequeue operations in circular queue. 8
- b) What is the advantage of doubly linked list over singly linked list? Write algorithms to insert and delete a node at the beginning of singly linked list 7
3. a) Write a simple program in C or C++ to create a singly linked list containing three nodes. The first node stores 11, second node stores 22 and third node stores 33. Also display the data of nodes. 8
- b) Write the cases in which you will use iteration and recursion. Explain how you will solve the Tower of Hanoi problem. 7
4. a) Why do you need a balanced tree? Create an AVL tree from the given set of values: 5, 7, 13, 9, 6, 3, 14, 10, 4. 8
- b) What is the advantage of variable length encoding? Construct Huffman tree and generate code for the following symbol with their frequency given below. 7

Symbol	C	D	E	K	L	M	U	Z
Frequency	32	42	120	7	42	24	37	2

5. a) Explain how you use divide and conquer strategy to sort the following list using quick sort : 7, 9, 45, 12, 56, 90, 3, 8, 50. 8
- Also write the best, average and worst case time complexity of quick sort for simple partition case.
- b) What is the purpose of hashing? For given input {2, 17, 49, 37, 25, 67, 82, 4, 9, 3} and a hash function $h(x) = x \bmod 10$ show the resulting: 7
 - i. Hash table using linear probing
 - ii. Hash table using chaining
6. a) What is directed and undirected graph? Explain adjacency matrix and adjacency list representation of graph, with suitable example. 7
- b) Write Dijkstra's Algorithm. Find the shortest path from P to U using Dijkstra's algorithm. 8
7. Write short notes on: (Any two) 2x5
 - a) Binary search
 - b) Kruskal's Algorithm
 - c) Big O Notation



POKHARA UNIVERSITY

Level: Bachelor

Semester – Spring

Year: 2020

Program: BE

Full Marks: 70

Course: Data Structure and Algorithm

Pass Marks: 31.5

Time: 2 hrs.

Candidates are required to answer in their own words as far as practicable.
The figures in the margin indicate full marks.

Attempt all the questions.

Group - A: (5×10=50)

- Q. N. 1 Justify the statement "Data Structure is the backbone of software programming".
Convert the given expression into Postfix expression showing the content of stack at each stack. $(A+B \times C/D)+E \times F-G \times H+I-J)$ 4+6

- Q. N. 2 Mention and judge partition algorithm of recursive Merge sort and Quick sort. 10

OR

- Q. N. 3 Construct a BST from the following data and show VLR, LVR, LRV and RVL traversals 14, 10, 17, 12, 11, 20, 18, 25, 20, 8, 22, 23. 10

- Q. N. 4 Differentiate between a perfectly balanced tree and AVL tree. Detect the effects of an insertion of a node on both the trees. 10

- Q. N. 5 Is it true that we can implement Stack and Queue using static and Dynamic memory allocation? Justify your understanding. Judge efficiently which of the memory allocation techniques are better in terms of contiguous list and linked list. 6+4

- Q. N. 6 Consider a table size 11 and elements are 28, 18, 23, 10, 36, 23, 40, 43 and hash function is $H(x)=x \bmod 11$. Find separate chaining, linear probing and quadratic probing. Why rehashing is needed? Explain with example.

Group - B: (1×20=20)

- Q. N. 6 a) Define graph and diagram?

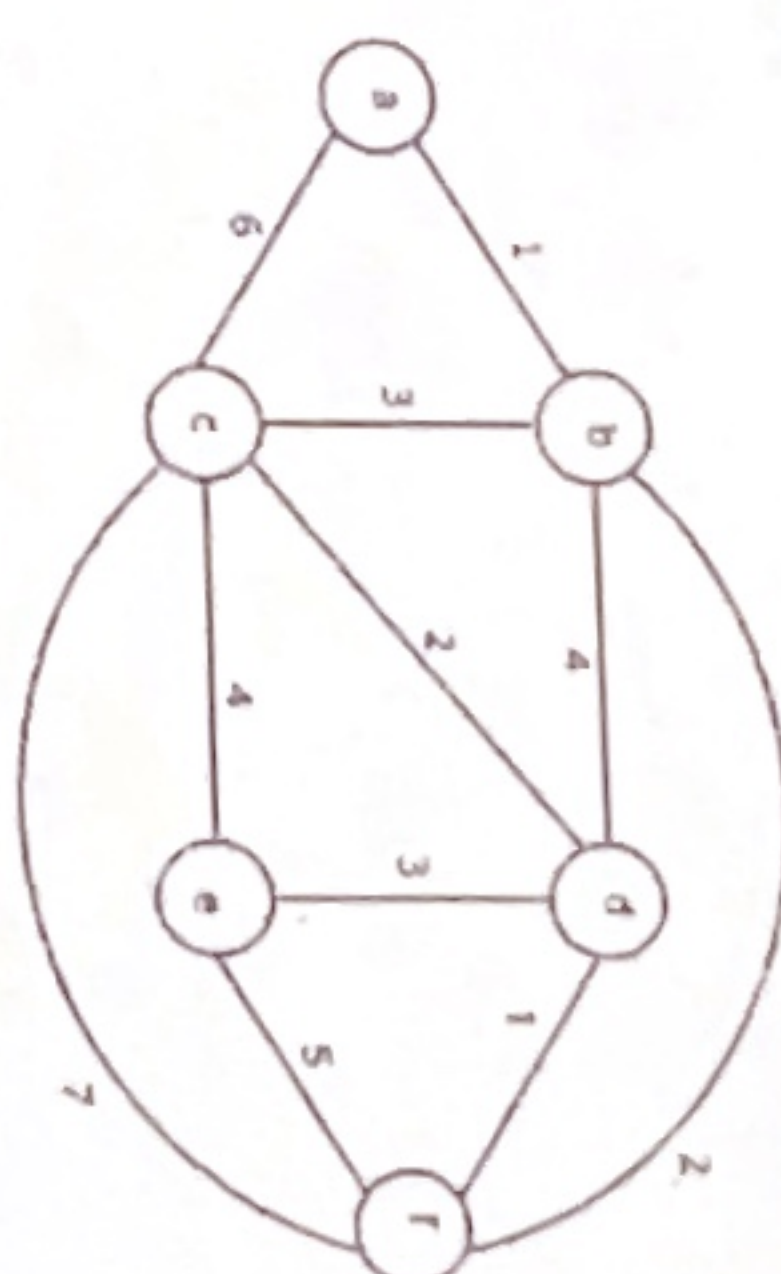
10

Let G be the graph represented by this adjacently list

Vertex	Adjacent list
A	F
B	C
C	B
D	A,B
E	C,D,
F	E

- i) Draw G
ii) Is G a directed graph?

- iii) Is G weakly connected?
iv) Give the adjacency matrix for G
b) Write Dijkstra's algorithm to find the shortest path of the following diagram from a to f.



POKHARA UNIVERSITY

Level: Bachelor
Semester: Fall
Programme: BE
Course: Data Structure and Algorithms

Year : 2021
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

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The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define Abstract Data type with example. Explain which data structure are suitable for the following problems with proper reasons
 - i. Evaluating the arithmetic expression
 - ii. Process scheduling by operating systems.
 - iii. Developing the social networks.
- b) Write the advantages of Postfix expression over the Infix expression, while processing by computer system. Convert the given expression into Postfix expression showing the content of stack at each step.
 $(A+B*(C/D))+E*(F-(G*H+I-J))$
2. a) Write algorithms to en-queue and de-queue in a circular queue.
- b) What is the advantage of an array over linked list and what is the advantage of linked list over an array? Implement a simple singly linked list with three nodes containing data 11, 22 and 33 using C or C++ code.
3. a) Write algorithms for push and pop operations on a stack using linked list implementation.
- b) Write algorithms to insert and delete a node at the end of singly linked list.
4. a) Explain the necessity of balancing a tree. How are the four different rotations applied to rebalance the AVL tree? Explain with example.
- b) Generate the Huffman Code for the following character with the given frequency.

Character	a	b	c	d	e
Frequency	8	6	3	4	15

- a) Explain the insertion sort with an appropriate example. Also state its time complexity.
- b) What is collision in hashing? What are the collision resolving techniques in hashing? Explain about separate chaining.
- a) Define graph and digraphs. Explain Adjacency matrix representation of graph with examples in undirected and directed graph.
- b) Define the minimum spanning tree. Explain Kruskal's algorithm for finding the minimum spanning tree.

Write short notes on: (Any two)

2×5

- a) TOH problem
- b) Divide and conquer algorithm
- c) Big O Notation

POKHARA UNIVERSITY

Level: Bachelor	Semester: Spring	Year : 2021
Programme: BE		Full Marks: 100
Course: Data Structure and Algorithm		Pass Marks: 45
		Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Explain the role of Data Structure in computer programming. Explain how you can represent a stack as an ADT. 8
 b) What are the advantages of postfix expression over infix expression? Write an algorithm to convert postfix expression to prefix expression. Convert the following postfix expression into prefix expression AB+CD* 7
2. a) Explain the application area of queue. What will happen if front is equal to rear in linear queue? Demonstrate the concept of enqueue, dequeue and traverse operation in a linear queue with suitable example and supporting algorithm. 8
 b) Differentiate between singly circular linked list and singly linked list. Implement queue using Linked list. 7
3. a) How are the higher order polynomials represented using linked list? With necessary supporting figures, write an algorithm to insert a node in given position of doubly linked list. 8
 b) In which conditions recursive algorithms are suitable? Explain with the problem of printing Fibonacci series. 7
4. a) Construct an AVL tree by inserting the following data: 7
 14, 17, 11, 7, 53, 4, 13, 12, 8, 60.
 Also delete the item 8, 7, 14, 17 from the constructed tree showing every step.
 b) What is B-Tree? What are the features of B-Tree? Discuss on the importance of B-Tree. 8

5. a) Sort the following data using quick sort explaining every step: 21, 43, 51, 32, 20, 35, 8, 12
 b) What is the principle of hash function in searching? For the data {1, 16, 49, 25, 64, 0, 81, 4, 9} and a hash function $h(x) = x \text{ mod } 10$, find the resulting:
 i) Hash Table using linear and quadratic probing.
 ii) Hash table using chaining.
6. a) What is a graph? Mention any two applications of graph in real world examples.
 b) Define graph traversal. Differentiate between DFS and BFS with an example.
7. Write short notes on: (Any two)
 a) Priority Queue
 b) Bubble Sort
 c) Big O notation

POKHARA UNIVERSITY

Level: Bachelor
Semester: Fall
Programme: BE
Course: Data Structure and Algorithm

Year : 2022
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

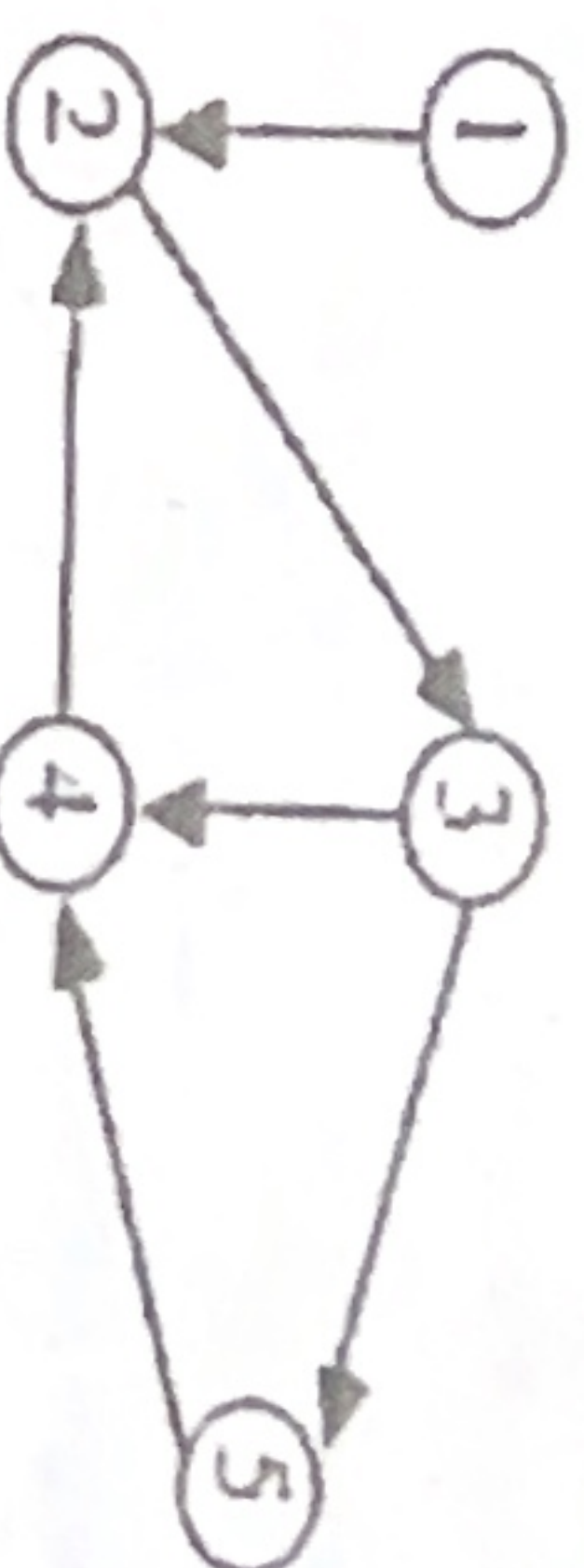
Attempt all the questions.

1. a) What is the role of data structures for algorithms? Create an ADT for Stack using C or C++ code. 7
b) Why do you need to convert an infix expression to postfix? How do you use stack to convert the following expression to postfix notation? $(A+B)*C-(D-E)/F$ 8
2. a) How does circular queue solve the problem of a linear queue? Implement the enqueue and dequeue operations in circular queue using C or C++ code. 8
b) In which condition you use linked list implementation of a queue? Explain with example. 7
3. a) Write an algorithm to insert a node at the beginning of a singly linked list. Also illustrate with an example. 7
b) What is the advantage of using recursive algorithms? Implement the recursive algorithms to solve the Tower of Hanoi problems using C or C++ code. 8
4. a) Construct an AVL tree in which elements are inserted in the following order. 50, 72, 96, 94, 107, 26, 12, 11, 9, 2. Show how the tree would look after the deletion of 26, 50, 16, and 10 respectively. Explain each steps of deletion. 8
b) Why do you need to balance binary search tree? Generate the Huffman code for the following character with the given frequency: 7

Character	K	O	P	R	H	A
Frequency	20	9	17	12	1	7

- a) Use insertion-sort to sort the following data: 7
24, 11, 49, 35, 98, 72, 34, 44

- b) How is the divide and conquer strategy used to search an item using binary search? Explain with a suitable example. 8
a) What do you mean by transitive closure of a graph? Find the transitive closure of the given graph.



- b) What is the purpose of Kruskal's algorithm? Explain with a suitable example. 8
7. Write short notes on: (Any two) 2x5
a) Hash Table and Hash Function
b) Big O Notation
c) B Tree

Candidates are required to give their answers in their own words as far as practicable.

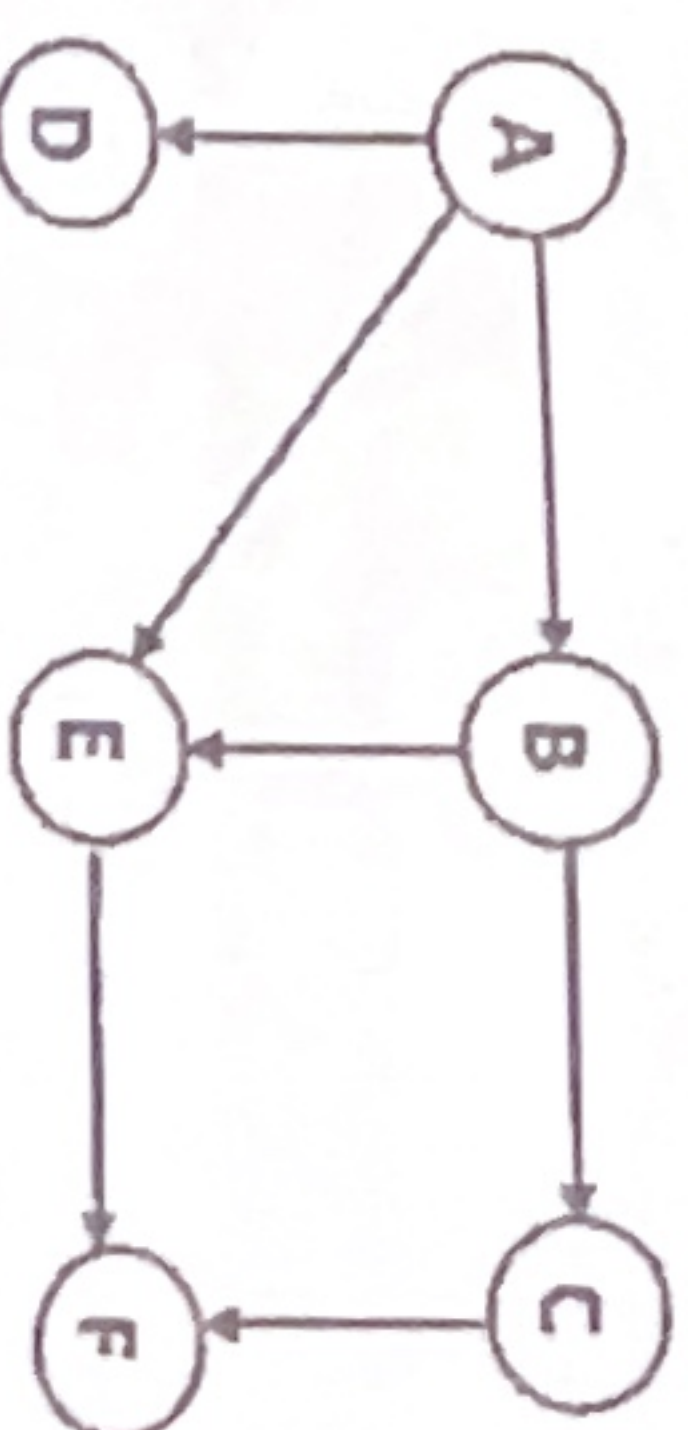
The figures in the margin indicate full marks.

Attempt all the questions.

- a) Why do you use Abstract Data Type (ADT)? Which data structure would you prefer to implement the following features? Justify with your reasons.
 - i. CPU Scheduling
 - ii. History on a Browser
 - iii. File Management System
 - iv. Display route on GPS
 - v. Representation of Polynomial equations
- b) Explain how you use stack in conversion of the following infix expression to a postfix expression using stack.
$$A + B - C / (D * F) - H * I$$
- a) What is a double ended queue? Explain a real world scenario in which the double ended queue can be applied.
- b) Write an algorithm to delete the last node in a singly circular linked list and also implement it using C or C++ code.
- a) What are the advantages and disadvantages of linked list implementation over array implementation? Explain the linked implementation of a stack
- b) How is a divide and conquer strategy used to recursively solve the Tower of Bhrahma (also called Tower of Hanoi)? Explain in detail. Also implement it using C or C++ code.
- i. How do you keep the cost of searching an item in an binary search tree to $O(\log n)$? Explain with an example.
- b) Construct a Huffman tree and generate Huffman codes for the following characters:

Character	C	D	E	K	L	M	U
Frequency	32	42	120	7	42	24	37

- a) Apply the radix sort algorithm to sort the numbers: 712, 205, 632, 377, 149, 877, 432, 317, 44 and 599
-) For the given data: 1, 16, 49, 36, 25, 64, 0, 81, 4, 9 and hash function:
 $h(x) = x \bmod 10$.
- i) Build a Hash table using quadratic probing
 - ii) Build a Hash table using chaining.
- a) Perform the Breadth First Search (BFS) on the following Graph explaining each step in detail.



- b) What is Dijkstra's algorithm? Explain the algorithm with a suitable example.

Write short notes on: (Any two)

- a) Priority Queue
- b) Doubly linked list
- c) Asymptotic Notation

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Data Structure and Algorithm (Old)

Semester: Fall
Year : 2023
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

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Attempt all the questions.

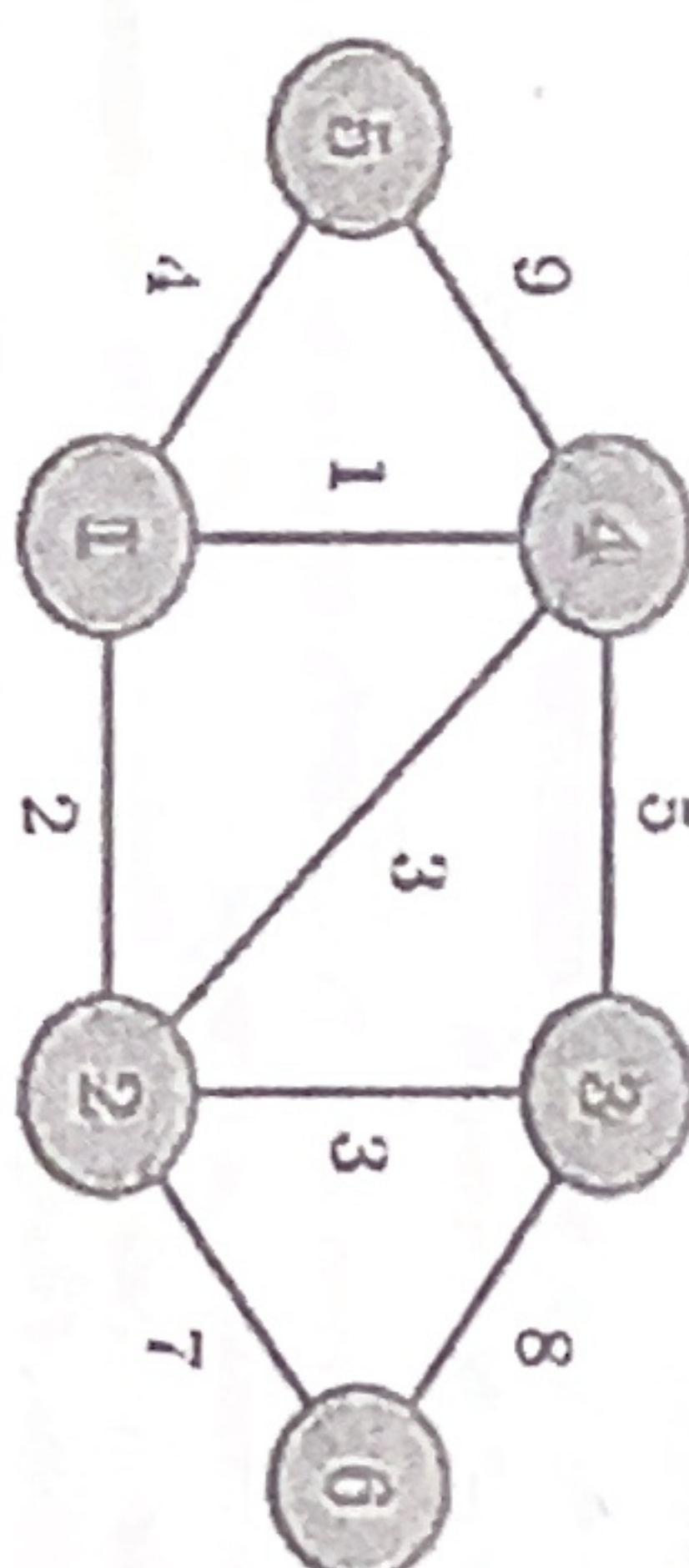
1. a) What are the differences between data types, data structures, and abstract data types (ADTs)? Explain their significance in programming with example. 7
- b) What is the principle of stack? Explain any three applications of stack in computer science with example. 8
2. a) Differentiate array with linked list. Write down an algorithm to perform the insertion at the beginning of a singly linked list. Also implement this algorithm using C++ code. 8
- b) What are the advantages of doubly linked list over singly linked list? Explain the linked implementation of queue. 7
3. a) What is balance factor in AVL tree? Construct an AVL tree from following sequence of number inserted in an order of empty binary tree. 20, 24, 29, 8, 3, 13, 27, 17, 14, 9, 1 8
- b) Why is balancing a tree important? Create a Huffman tree of a string: "rtletrergl". 7
4. a) Why is stable sorting important? Explain with the following data using radix sort: 13, 67, 27, 94, 63, 83, 94, 66, 32. 7
- b) Define hash system. Insert the elements {22, 91, 79, 34, 92, 41, 66, 86} using hashing function $key \bmod 10$ and resolve the collision with quadratic probing. 8

What is a graph? Draw graphs using the following adjacency matrices: 8

i. $\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$

ii. $\begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}$

- b) Define transitive closure. Write down an algorithm to perform the DFS traversal on a graph. 7
6. a) What is minimum spanning tree? Perform Dijkstra's algorithm to find the shortest path from vertex 5 to any other vertex for following graph. 8



- b) Explain how divide and conquer strategy is used to design an algorithm with a suitable example. 7
7. Write short notes on: (Any two) 2x5
 - a) Priority queue
 - b) Principle of recursion
 - c) Binary search algorithm

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Data Structure and Algorithms

Semester: Spring

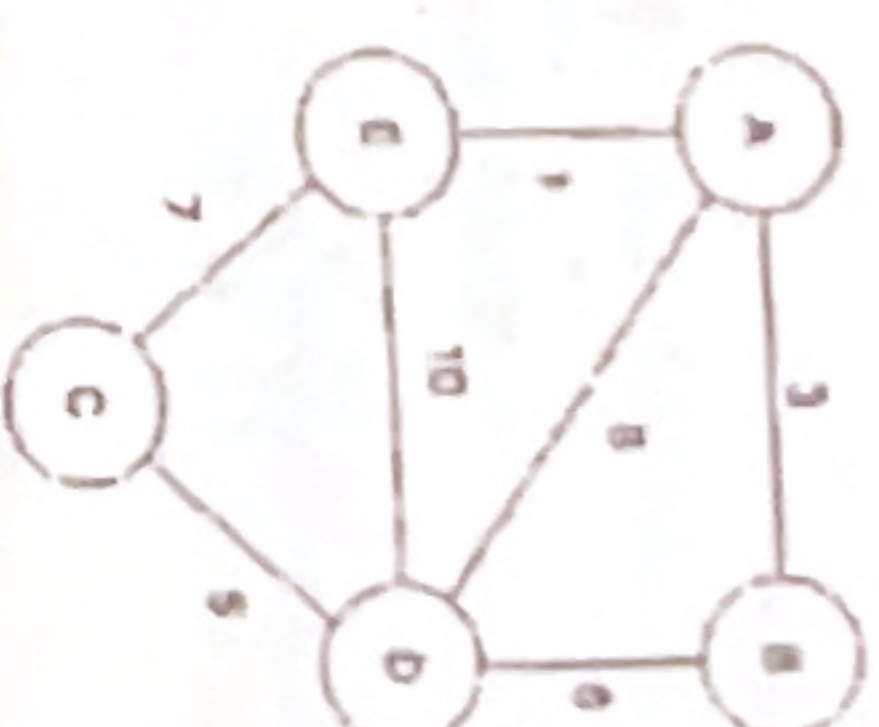
Year : 2024
Full Marks: 100
Pass Marks: 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

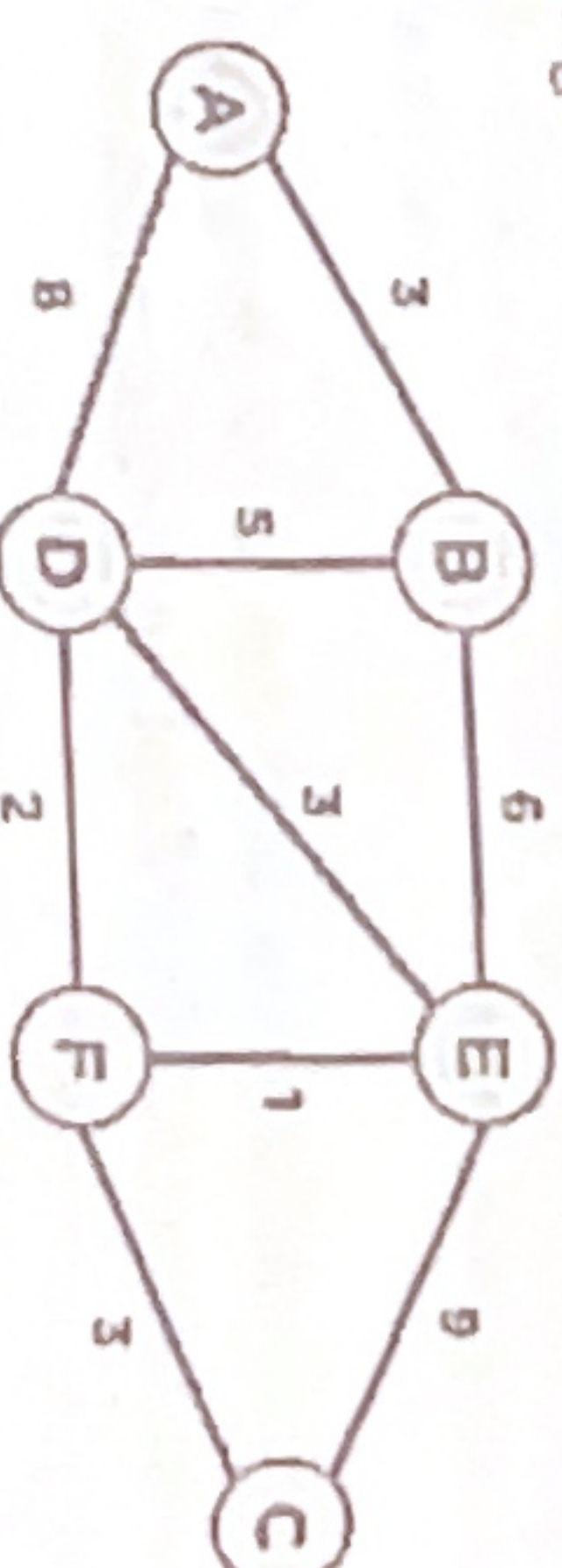
The figures in the margin indicate full marks.

Attempt all the questions.

- Define Abstract Data Type (ADT). Explain how ADTs provide data abstraction with an example. 7
 - Define a stack ADT. Explain and write algorithms for basic operations of stack. 8
- Define priority queue. Write C/C++ functions to implement the operations in a circular queue. 8
 - Explain List ADT. Write down an algorithm to delete the last node in a singly circular linked list. Explain each steps in detail. 7
- Write an algorithm to delete a node from the specific position of doubly linked list. 7
 - What is a linked stack? Implement its push and pop operations using singly linked list. 8
- How do you use recursion to solve a problem? Explain in detail with your own example. 7
 - Define a binary tree. Write algorithms for insertion and deletion operation in a complete binary tree. 8
- Explain the significance of a balanced tree over unbalanced tree. 8
 - Construct an AVL tree from the following data: 10, 20, 30, 15, 5, 12, 25, 50. 8
- Why do you think sorting is necessary? Sort the following data using quick sort: 5, 3, 8, 6, 2, 7, 1, 4 7
 - Why spanning tree is important? Find minimum spanning tree of following graph using Prim's algorithm: 7



- Find the shortest path from source vertex A to other vertices using Dijkstra's algorithm for the following graph.



- Write short notes on: (Any two)
- Binary search
 - Big O Nation
 - Hash System

POKHARA UNIVERSITY

Level: Bachelor
Semester: Spring
Programme: BE
Course: Data Structure and Algorithms

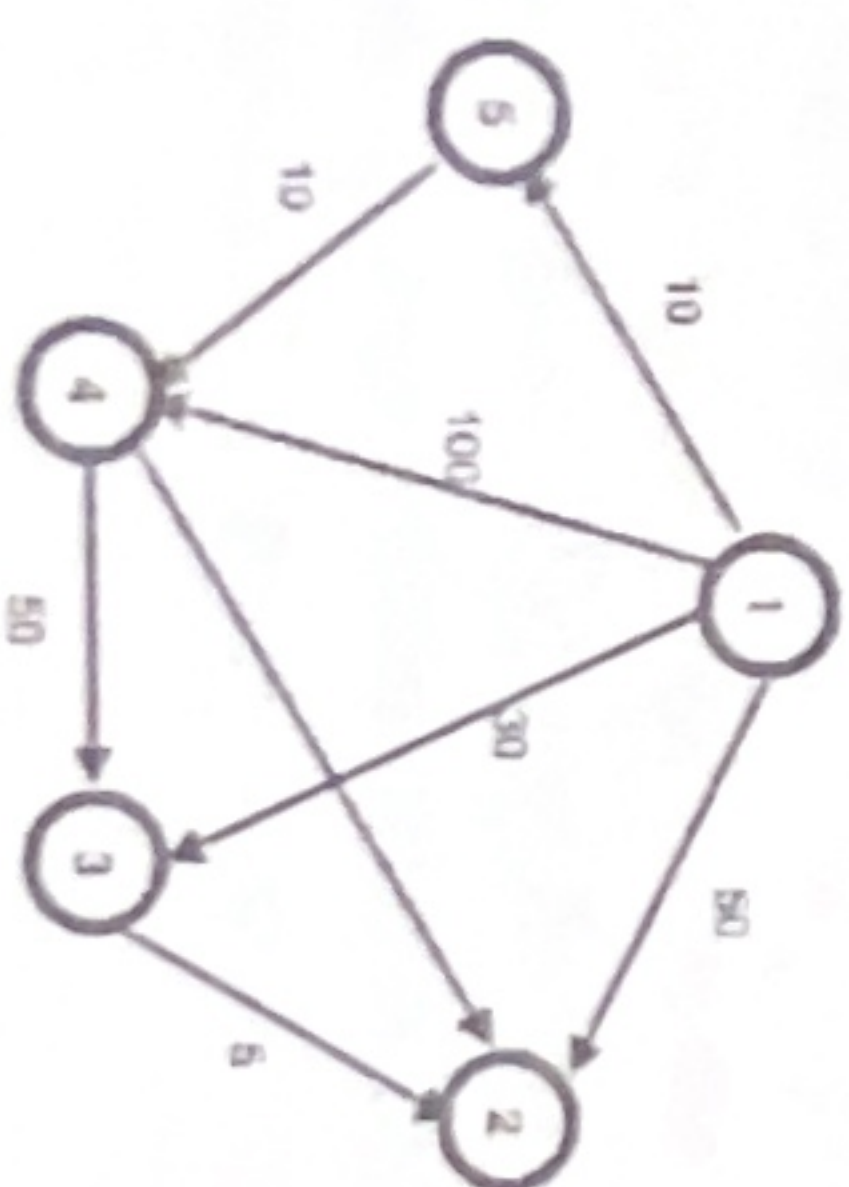
Year : 2025
Full Marks : 100
Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

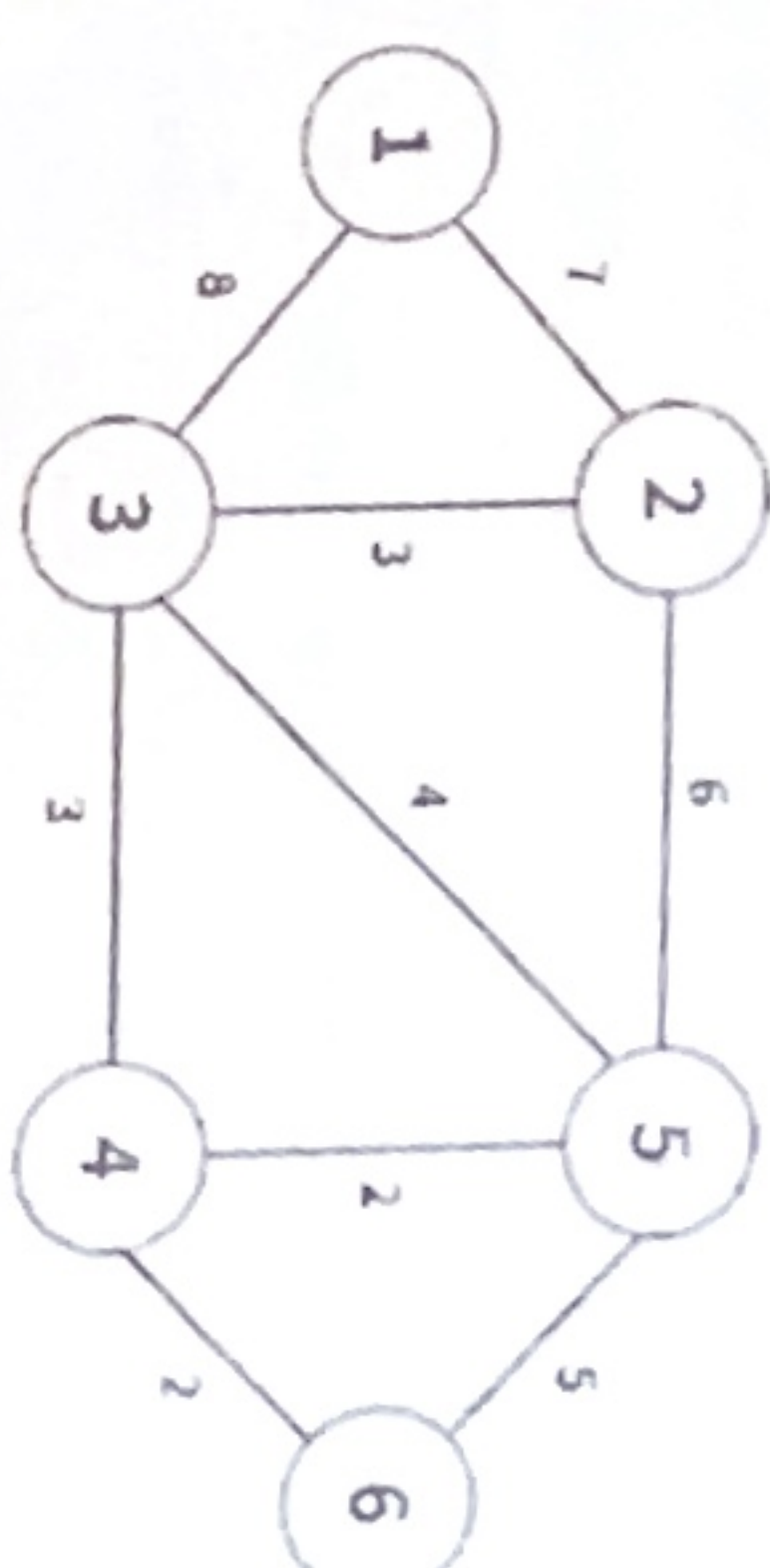
The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What is an ADT? How the knowledge of Data Structure can be used to develop real life applications? 4+4
- b) Convert the following infix expression to postfix $A + (B * C - (D / E \wedge F - H) * G) * H$ using stack. 7
2. a) Define queue. Write a C/C++ program to implement enqueue and dequeue in linear queue. 8
- b) Define singly linked list. Explain structure of Node/Object that represent linked list element. Write pseudo code to delete first node from the linked list. 1+3+3
3. a) What are the drawbacks of linear queue? Write algorithms to insert and delete an item in the circular queue. 2+6
- b) What is the use of recursion? Write a program for Tower of Hanoi. 7
4. a) Why do we need to balance the tree? Create an AVL tree from following keys 8,2,1,4,5,9,7,16,15,14,13,22,11. 8
- b) Construct the Huffman tree for "DATAWESOME" generating the Huffman code. 7
5. a) Sort the following numbers using the quick sort 40,60,10,20,50,30. 8
- b) Explain application areas of hashing. What are collisions and how they could be addressed? 7
6. a) Find the shortest path from source vertex 5 to other vertices using Dijkstra's algorithm. 8



- b) Define graph, connected graph and spanning tree. Find MST using Kruskal Algorithm. 7



Graph G

7. Write short notes on: (Any two) 2×5
 - a) Radix sort
 - b) Application of Stack
 - c) Asymptotic Notation